



iCET Certified Developer

COURSEWORK

Module	Data Base Management System
Coursework Title	Final Coursework
Coursework Type	Practical Exercise

1. Scenario - CityMed Hospital

CityMed is a busy urban hospital with multiple departments, wards, doctors, nurses, and thousands of patient records. The hospital currently manages all its data using paper records and spreadsheets. Your task is to design and implement a relational database that digitises this system.

You have been hired as the database developer. Study the information below carefully it describes all the data the hospital needs to store and the rules that govern it.

1.1 Hospital Departments & Wards

CityMed is divided into departments (e.g. Cardiology, Orthopaedics, Neurology, Paediatrics, General Surgery). Each department has a name, a physical location within the hospital building, a phone extension, and the name of the head doctor.

Within each department there are one or more wards. A ward has a name, a maximum patient capacity (between 1 and 100), and a ward type (General, ICU, Surgical, Maternity, or Paediatric).

1.2 Medical Staff

Doctors are employed by the hospital and belong to exactly one department. Each doctor has a first name, last name, speciality, email address (unique), phone number, hire date, and salary. Salaries must always be positive.

Nurses are assigned to a specific ward. Each nurse has a first name, last name, email (unique), and a shift (Morning, Evening, or Night).

1.3 Patients

When a patient visits CityMed for the first time they are registered with: first name, last name, date of birth, gender (Male / Female / Other), blood group, phone number, optional email, optional address, and the date they registered.

1.4 Appointments

A patient may book many appointments, but each appointment is with exactly one doctor. An appointment records the date, time, the reason for the visit, its status (Scheduled, Completed, Cancelled, No-Show), and optional clinical notes.

1.5 Prescriptions

After a completed appointment, a doctor may prescribe one or more medicines. Each prescription line records the medicine name, dosage, duration in days (1–365), and any special instructions. The same medicine cannot be prescribed twice in the same appointment.

1.6 Billing

Every appointment generates at most one bill. A bill records the total amount charged, the amount already paid, the payment status (Unpaid, Partial, or Paid), and the billing date. The amount paid can never exceed the total amount.

Schema Info

departments —< wards (one department has many wards)
 departments —< doctors (one department has many doctors)
 wards —< nurses (one ward has many nurses)
 patients —< appointments (one patient has many appointments)
 doctors —< appointments (one doctor has many appointments)
 appointments —< prescriptions (one appointment has many prescription lines)
 appointments —o bills (one appointment has at most one bill)

2. DDL Tasks - Data Definition Language

In this section you will create the database and all its tables, then make changes to the schema using ALTER and DROP statements.

2.1 Create the Database

Q1. Write the SQL statement to create a database named citymed_hms using the utf8mb4 character set and select it for use.

2.2 Create Tables

Create the following eight tables. Each table specification describes the columns required and the constraints that must be enforced. Choose appropriate data types.

Task 2.2.1 - departments

Column	Constraint / Rule
dept_id	Primary key, auto-increment
dept_name	NOT NULL, must be unique
location	NOT NULL
phone_ext	NOT NULL, exactly 10 characters
head_doctor	Optional (nullable)

Q2. Write the CREATE TABLE statement for departments.

Task 2.2.2 - wards

Column	Constraint / Rule
ward_id	Primary key, auto-increment
ward_name	NOT NULL
dept_id	NOT NULL, foreign key → departments
capacity	NOT NULL, must be between 1 and 100 (CHECK constraint)
ward_type	ENUM: General, ICU, Surgical, Maternity, Paediatric - default General

Q3. Write the CREATE TABLE statement for wards including the foreign key and CHECK constraint.

Task 2.2.3 - doctors

Column	Constraint / Rule
doctor_id	Primary key, auto-increment
first_name	NOT NULL
last_name	NOT NULL
speciality	NOT NULL
dept_id	NOT NULL, foreign key → departments
email	NOT NULL, unique
phone	NOT NULL
hire_date	NOT NULL, DATE type
salary	NOT NULL, DECIMAL(10,2), must be ≥ 0

Q4. Write the CREATE TABLE statement for doctors.

Task 2.2.4 - nurses

Column	Constraint / Rule
nurse_id	Primary key, auto-increment
first_name	NOT NULL
last_name	NOT NULL
ward_id	NOT NULL, foreign key → wards
shift	ENUM: Morning, Evening, Night - NOT NULL
email	NOT NULL, unique

Q5. Write the CREATE TABLE statement for nurses.

Task 2.2.5 - patients

Column	Constraint / Rule
patient_id	Primary key, auto-increment
first_name	NOT NULL
last_name	NOT NULL
dob	NOT NULL, DATE
gender	ENUM: Male, Female, Other - NOT NULL
blood_group	ENUM: A+, A-, B+, B-, AB+, AB-, O+, O- – nullable

Column	Constraint / Rule
phone	NOT NULL
email	Nullable
address	Nullable, TEXT
registered_on	NOT NULL, DATE, default today's date

Q6. Write the CREATE TABLE statement for patients.

Task 2.2.6 - appointments

Column	Constraint / Rule
appointment_id	Primary key, auto-increment
patient_id	NOT NULL, foreign key → patients
doctor_id	NOT NULL, foreign key → doctors
appt_date	NOT NULL, DATE
appt_time	NOT NULL, TIME
reason	NOT NULL, VARCHAR(255)
status	ENUM: Scheduled, Completed, Cancelled, No-Show - default Scheduled
notes	Nullable, TEXT

Q7. Write the CREATE TABLE statement for appointments.

Task 2.2.7 - prescriptions

Column	Constraint / Rule
prescription_id	Primary key, auto-increment
appointment_id	NOT NULL, foreign key → appointments (CASCADE on delete/update)
medicine_name	NOT NULL, VARCHAR(150)
dosage	NOT NULL
duration_days	NOT NULL, unsigned, CHECK 1-365
instructions	Nullable, TEXT
issued_date	NOT NULL, DATE, default today

Unique rule: The same medicine cannot be prescribed more than once per appointment. Add a composite UNIQUE constraint on (appointment_id, medicine_name).

Q8. Write the CREATE TABLE statement for prescriptions.

Task 2.2.8 - bills

Column	Constraint / Rule
bill_id	Primary key, auto-increment
appointment_id	NOT NULL, unique, foreign key → appointments
total_amount	NOT NULL, DECIMAL(10,2), >= 0
paid_amount	NOT NULL, DECIMAL(10,2), >= 0, default 0.00
payment_status	ENUM: Unpaid, Partial, Paid - default Unpaid
billing_date	NOT NULL, DATE, default today

Extra rule: paid_amount must never exceed total_amount. Enforce this with a CHECK constraint.

Q9. Write the CREATE TABLE statement for bills including both CHECK constraints.

2.3 ALTER TABLE

Q10. A new requirement has arrived: the patients table must store an emergency_contact field (VARCHAR, nullable). Write the ALTER TABLE statement to add this column after the address column.

Q11. The phone column in the doctors table needs to be extended from VARCHAR(20) to VARCHAR(30) to support international numbers. Write the ALTER TABLE statement.

Q12. Write an ALTER TABLE statement to add a named CHECK constraint to the appointments table that prevents an appointment from being booked on a past date (appt_date must be >= today).

2.4 DROP

Q13. Write the DROP TABLE statements to remove all eight tables in the correct order so that foreign key constraints are not violated. Explain in a comment why the order matters.

3. DML Tasks - Data Manipulation Language

3.1 INSERT

Populate the database with realistic sample data. All inserts must respect the constraints you defined in Section 2.

- Q14.** Insert at least 4 departments into the departments table.
- Q15.** Insert at least 2 wards per department (minimum 6 wards total).
- Q16.** Insert at least 5 doctors spread across different departments.
- Q17.** Insert at least 3 nurses assigned to different wards.
- Q18.** Insert at least 6 patients with varied genders and blood groups.
- Q19.** Insert at least 8 appointments. Include a mix of statuses: Scheduled, Completed, Cancelled.
- Q20.** Insert prescriptions for at least 4 completed appointments (at least one appointment should have two medicines).
- Q21.** Insert bills for each appointment that has a prescription. At least one bill should be 'Partial' and one 'Unpaid'.

3.2 UPDATE

- Q22.** One appointment needs to be marked as 'Cancelled' and a cancellation note added to its notes field. Write the UPDATE statement.
- Q23.** A patient has made an additional payment of 5,000 toward their bill. Write an UPDATE that increases paid_amount by 5,000 and automatically sets payment_status to 'Paid' if the full amount is now covered, or keeps it 'Partial' otherwise. Use a CASE expression.
- Q24.** All doctors in the Cardiology department are to receive a 10% salary increase. Write an UPDATE that uses a JOIN between doctors and departments to apply this change.

3.3 DELETE

- Q25.** Write a DELETE statement that removes all bills whose linked appointment has a status of 'Cancelled'. Use a subquery.
- Q26.** Write a DELETE statement that removes all 'No-Show' appointments that are older than 2 years.

4. Query & JOIN Tasks

Q27. List all patients showing: patient_id, full name (first and last combined as full_name), date of birth, gender, blood group, and phone. Order alphabetically by last name.

Q28. List all upcoming appointments (status = 'Scheduled', date >= today) showing the patient name, doctor name, appointment date, time, and reason. Order by date and time.

Q29. Write a query using INNER JOIN to display full appointment details: appointment_id, patient name, doctor name, department name, appointment date, time, and status.

Q30. Using INNER JOIN, list each prescription showing: prescription_id, patient name, doctor name, medicine name, dosage, and duration. Only include completed appointments.

Q31. Using a LEFT JOIN, display every patient along with the total number of appointments they have had (including patients with zero appointments). Label the count as total_appointments. Order by total_appointments descending.

Q32. Using a LEFT JOIN between wards and nurses, show every ward along with the count of nurses assigned to it. Include wards with no nurses assigned.

Q33. Using a RIGHT JOIN, list every doctor and the number of appointments they have. Include doctors who have no appointments yet. Label the count as appointment_count and order descending.

Q34. Write a query to produce a department revenue report showing: department name, total appointments, total billed amount (gross_revenue), total collected (collected), and outstanding balance. Group by department and order by gross_revenue descending.

Q35. Write a query using a subquery to list the name and phone of every patient who has at least one unpaid or partially paid bill.

Q36. Write a correlated subquery to find all doctors whose salary is above the average salary of their own department. Show doctor name, department, and salary.

5. Constraint Tasks

Answer the following questions about constraints in the CityMed HMS schema.

5.1 Identify & Explain

Q37. List all PRIMARY KEY constraints in your schema. For each, state the table name, the column(s) involved, and explain why that column was chosen as the primary key.

Q38. List all FOREIGN KEY constraints in your schema. For each, state the referencing table, the referenced table, and the ON DELETE / ON UPDATE behaviour you chose. Justify your choices.

Q39. Identify every UNIQUE constraint in your schema and explain the business rule each one enforces.

Q40. List every CHECK constraint you created. For each, write the constraint in plain English and explain what data error it prevents.

5.2 Apply

Q41. Write an ALTER TABLE statement to add a CHECK constraint to the prescriptions table that limits duration_days to a maximum of 365 days.

Q42. Doctors occasionally transfer between departments. Explain what happens to their department record when a department row is deleted, given the ON DELETE behaviour you chose for the fk_doctor_dept foreign key. Would you change that behaviour? Justify your answer.

Q43. A new business rule states that a bill's billing_date must not be earlier than its appointment date. Write the ALTER TABLE statement to add this constraint to the bills table.

5.3 Short Answer

Q44. What is the difference between a UNIQUE constraint and a PRIMARY KEY constraint? Give an example from your schema for each.

Q45. Explain what referential integrity means and describe one scenario in the CityMed system where a foreign key constraint protects referential integrity.

Student Guidelines for Coursework Submission: CityMed Hospital Database Management System

Please adhere to the following guidelines when completing and submitting your CityMed Hospital Database Management System assignment.

1. Submission Requirements

- **Document Format:** All answers, SQL code, and required evidence must be compiled into a single digital document, which should be a **Microsoft Word Document (.docx)** or a **Google Document**.
- **Completeness:** You must provide a clear and comprehensive answer for **all questions** listed in the coursework (Q1 through Q45).
- **Question Numbering:** Clearly label each answer with the corresponding question number (e.g., **Q1**, **Q27**, **Q44**).
- **Submission File Naming & Format:** Your final document must be saved as a PDF file with the naming convention: **DBMS_FINAL_CW_<Your_StudentID>**.
 - Example: **DBMS_FINAL_CW_PR26120345**
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2. Content Specifics

- **SQL Tasks (Sections 2, 3, 4):** For all DDL, DML, and Query questions, you must include the complete, correct, and final SQL code used to solve the problem directly in your answer document. Adhere strictly to the required formatting: SQL keywords in **UPPERCASE**, and table/column names in **lowercase**.
- **Constraint Tasks (Section 5):** Provide detailed, well-justified, and technically accurate explanations for all constraint-related questions.

3. Mandatory Evidence

A critical component of this submission is proving the correct implementation of your database schema and data integrity rules.

- **Database Structure Screenshot:** After successfully completing the DDL (Section 2) and DML (Section 3) tasks -meaning all eight tables are created and all sample data has been inserted-you must include a **screenshot** clearly showing the structure of the entire citymed_hms database.
 - **Requirement:** This screenshot must be taken from your SQL client (e.g., MySQL Workbench, DBeaver) and must clearly display the names of all eight tables, their columns, data types, and the definitions of Primary Keys, Foreign Keys, and other constraints. This should represent the database after all data has been loaded.

4. General Advice

- Read the entire scenario (Section 1) and all table specifications carefully before beginning to write your SQL.
- Ensure that all INSERT, UPDATE, and DELETE statements respect the constraints you have defined. Test your code thoroughly before adding it to your final submission document.